THE SI SENTINEL

Weekday briefing on builders shaping super intelligence

TUESDAY, NOVEMBER 3, 2026

THE MORNING PICTURE

Musk does the radiator math for orbital compute, a frontier model edits another model's weights, and Muse pays $300,000 for one bug

Elon Musk said cooling chips is "much easier" in space and put numbers on it: about 1,300 watts per square meter from an aluminum radiator, against about 250 from solar panels. MIT researchers had GPT-6 Astra edit 0.016% of a small model's weights and watched its math scores jump, and Tobi Lütke called it "exactly as crazy as it seems." Mistral published the technical report for its now open-weight Mistral Large 4.

On security, David Singleton said Muse has paid more than $1 million in bug bounties, and supermarket freezers in at least seven European countries were hacked. OpenAI described Pando, a new Git backend for its agent-heavy monorepo. David Heinemeier Hansson warned that agents will be hard on "cantankerous" programmers, and in Tokyo, Tim Sweeney defended Epic's AI-first plan for Unreal Engine 6.

I · ORBITAL COMPUTE

"Much easier to cool in space," Musk says, with numbers

A Sequoia partner described a passive radiator loop for orbital data centers. Musk replied with the heat budget and said super intelligence compute will "round up to 100%" in space.

~1,300	~250	~5X
Watts per square meter Musk says an aluminum radiator can shed facing away from the Sun	Watts per square meter solar panels generate, by his figures	Solar panel area he expects relative to radiator area

Elon Musk @elonmusk · CEO, SpaceX and Tesla

Shaun Maguire, a partner at Sequoia Capital with a physics doctorate from Caltech, argued Monday that once Starship can lift "huge amounts of mass into space cheaply, it is genuinely easier to cool things in space." His design is passive: an aluminum structure with a working fluid such as ammonia that boils near the chips and circulates heat to the radiator, which sheds it as black-body radiation. "Black body radiation goes like temperature to the 4th power so the hotter you can run chips and hotter you can get the working fluid the better you can shed heat," he wrote. Musk answered at 10:43pm EST:

> *"Yeah, much easier to cool in space. ~1300 W/m^2 is easily achievable for an aluminum radiator pointed away from the Sun, as it is mostly just seeing cosmic background radiation at ~3 degrees above absolute zero! Solar panels generate ~250 W/m^2, so you should expect them to have ~5 times the area of the radiator. Most of the mass of the satellite is solar panels."*
>
> ELON MUSK · @ELONMUSK · 10:43PM EST · POST

W/m^2 is watts per square meter. To a user who noted that radiative heat transfer "grows gigantically" at GPU temperatures, he added: "Yup, T^4 scales super fast."



The International Space Station in December 2002, after Space Shuttle Endeavour's crew installed the P1 truss, which added three radiators to the station's thermal control system. The white panels are radiators; the gold wings are solar arrays.

PHOTO: NASA (STS-113, S113-E-05448) · PUBLIC DOMAIN

Shaun Maguire and the long view

Earlier Monday, Maguire wrote that "SpaceX is the low-cost producer for compute (initially terrestrial which becomes orbital)" and predicted it can turn on new data centers "2-3 years faster than its competitors." Musk replied at 10:08am EST: "Future super intelligence compute will obviously be overwhelmingly in space. By 'overwhelmingly', I mean that it will round up to 100%!" When Maguire promised "a systems level analysis of orbital compute over the next week or two," saying "Almost all SI compute will be in space," Musk wrote "Absolutely." SI is super intelligence.

II · MODELS AND RESEARCH

A frontier model rewrites a small model's weights, and Mistral opens its largest model

MIT researchers call it Weight-Space Programming. Mistral released the report behind Mistral Large 4, and a fine-tune from Suhail Doshi's Particle reached a data-analytics frontier.

0.016%
Share of Qwen3.5-0.8B's weights GPT-6 Astra changed, with no training

7.9 → 48.1%
The edited model's MATH-500 score, before and after

40 of 43
Questions Particle-1 passed on Plotly's data-analytics exam

Tobi Lütke @tobi · CEO, Shopify

Rikiya Takehi, a first-year doctoral student at MIT's Computer Science and Artificial Intelligence Laboratory, posted Monday: "We asked GPT-6 Astra to edit Qwen3.5-0.8B weights to improve math. It directly changed only 0.016% of weights, fixing 0.8B's reasoning loops & improving math up to 6x and even instruction following 3x, without gradients or training." Lütke quoted it at 6:30pm EST: "Yea. this is exactly as crazy as it seems."

The paper, by Takehi, Jayden Teoh, Alex Zhang and Omar Khattab, says Astra traced Qwen's thinking loops to one direction in layers 15 and 19 and removed it from 131,072 weights. MATH-500 rose from 7.9% to 48.1%, GSM8K from 13.8% to 67.1% and the IFEval instruction-following test from 21.0% to 62.3%. Astra also identified what small networks were trained to do just by reading their raw weights. Tim Sweeney (@TimSweeneyEpic), chief executive of Epic Games, replied to a stunned reader: "Next phase, they'll just read the binary file and 'see the matrix' as Neo did and we other humans cannot."

Guillaume Lample @GuillaumeLample · Co-founder and chief scientist, Mistral AI

Pierre Stock, Mistral's vice president of science, wrote: "ML4 is now open-weight! Frontier capabilities on code agents & cyber, way more efficient and compact than its Preview version three weeks ago." Lample quoted him at 1:30pm EST: "We just released a technical report on the training of ML4." Mistral Large 4 is a 1-trillion-parameter mixture-of-experts model with about 52 billion parameters active per token and a 1-million-token context, trained on 3,800 Nvidia Grace Blackwell chips in Europe, its model card says. Mistral's announcement puts it at 48% on the Artificial Analysis Coding Agent Index, effectively tied with GLM-5.3 while using 47% fewer tokens, and 41% on Terminal-Bench 4.0.

Suhail Doshi @Suhail · Founder, Particle

Plotly's co-founder wrote that Particle's model "landed on our pareto frontier for data analytics tasks, beating out Terra-6.1 and Haiku 5.5. First time a new player has landed on our frontier in years." Doshi quoted it at 3:10pm EST: "Pareto frontier on data analytics. Honestly feels like a full circle moment." He founded the analytics company Mixpanel. Plotly's write-up says Particle-1, a reinforcement-learning fine-tune of DeepSeek V4.1 Flash, passed 40 of 43 questions for $0.35, tying GPT-6.1 Terra for about a quarter of the cost. Asked about cyber models, Doshi replied "Soon."

> QUICK HIT
>
> **François Chollet** (@fchollet), co-founder of Ndea and creator of Keras, at 12:59pm EST Monday: "The most grating aspect of LLM writing is that it tries to turn every single sentence into a punchline, a witty slogan, with no regard for which sentences need emphasis and which ones don't." LLM stands for large language model.

ON THE WIRE

Cheaper frontier. Particle-1 tied GPT-6.1 Terra on Plotly's exam at $0.35 a run, about a quarter of the cost, per Plotly.

III · WORK AND AGENTS

Agents raise the ceiling, and the people who manage them stay accountable

Levie said the gap between frontier users and everyone else has never been wider. Heinemeier Hansson and Uncle Bob Martin talked about who thrives and who answers for the work.

30–50%
Best productivity gains from AI a few years ago, by Levie's estimate

5–10X
The upper limit now, "depending on the field," he wrote

3
Round-the-clock agents Doshi plans: security, product data and personal

David Heinemeier Hansson @dhh · Creator, Ruby on Rails and Omarchy

> *"The age of agents is going to be amazing for most programmers, but it'll be hard times for the cantankerous ones. Specialized knowledge and rare skills will no longer offer immunity for those who are a pain to work with. Best learn some basic people skills."*
> DAVID HEINEMEIER HANSSON · @DHH · 8:03AM EST · POST

Quoting a user who argued that rudeness to agents carries over to colleagues, he wrote: "How you do anything is how you do everything." Engineer Geoffrey Huntley said open-source codes of conduct "need to be retired; I deeply regret championing them." Heinemeier Hansson at 11:19am EST: "Amen. We never needed to codify basic decency, but in doing so created legalisticl weapons of mass coercion and usurpation. Never again."

Uncle Bob Martin @unclebobmartin · Author, Clean Code

Asked who checks the architecture when every engineer manages language models, Martin wrote at 12:44pm EST: "The engineers check the architecture. All accountability lies with the engineers. The agents are completely unaccountable." To a user who said humans will never be as smart as AI, he replied early Tuesday: "We cannot out-reason it. But we CAN out-create and out-discern it. And, in the end, we must take responsibility for everything we allow it to do."

Aaron Levie @levie · CEO, Box

Levie wrote at 1:34am EST that "the divergence in AI output weirdly may be the highest it's ever been, even though adoption is the broadest it's ever been." A few years ago, a very good user in coding might get "maybe 30-50% better productivity." Now models run far longer tasks with little human help, and "the upper limits on productivity gains have gone from 30-50% to now likely 5-10X depending on the field." He expects the gap to widen "given the rate of progress we're likely to see with RSI," or recursive self-improvement.

Suhail Doshi @Suhail · Founder, Particle

At 8:07pm EST: "In a year, we will all have multiple 24/7 agents running for all facets of our personal and working lives—working proactively on our behalf. The sheer quantity of tokens computed will be staggering." His own would cover security, product data analysis and his personal life, he added, and "All companies will be forced to do this to compete at the same velocity."

QUICK HIT
Paul Graham (@paulg), co-founder of Y Combinator, at 6:55am EST Tuesday, replying to a prediction that the AI bubble will burst: "You can't claim both that AI is fake and that it's destroying jobs. If it were fake it wouldn't be able to."

IV · SECURITY AND INFRASTRUCTURE

Muse's bug bounty passes $1 million as hackers turn up Europe's supermarket freezers

Singleton laid out the worst bug Muse has paid for. OpenAI rebuilt its Git server for agents, and Vercel and Suleyman each pressed on safety.

$300,000
Muse's largest single bounty, for an authentication bug found before launch

64
Valid findings among more than 630 Muse bounty submissions

~80%
Cut in push latency after OpenAI moved its monorepo to Pando

David Singleton @dps · Meta Superintelligence Labs

Singleton, the former chief technology officer of Stripe, posted at 9:59pm EST: "We've now paid out more than $1M in @muse bug bounties. That includes $300,000 to a security researcher who found the most critical bug we've seen in Muse." The program has covered Muse privately since April and publicly

since its Sept. 8 launch, with "630+ submissions, 64 valid findings and ~$1.2M awarded so far," he wrote.

The $300,000 bug, found four months before launch, was an authentication check that "trusted a header added by a local proxy as proof a request was safe," he explained. A migration added a second proxy layer, so outside requests could carry that header too, exposing testers' tokens and conversations in pre-release virtual machines. Muse "shut down the whole service."

> *"Security reviews ask "did we build what we said?" Attackers ask "what did you forget, and what can I chain together to break in?" Any system that claims to be secure needs adversarial testing, and bug bounties are one of the best ways to do it at scale."*
>
> DAVID SINGLETON · @DPS · 9:59PM EST · POST

The program pays "up to $300K per report, including up to $130K for a prompt injection against a single user."

Elon Musk @elonmusk · CEO, SpaceX and Tesla

Supermarket chains in Belgium, Estonia, Finland, Ireland, Luxembourg, the Netherlands and Sweden reported Monday that hackers had raised refrigerator and freezer temperatures, some to 50°C (122°F), The Guardian reported, echoing an October attack in France. Musk, at 10:13am EST: "Literally an episode of Silicon Valley."



Building 92, the visitor center on Microsoft's campus in Redmond, Washington, in June 2015.
PHOTO: JIAQIAN AIRPLANEFAN · CC BY 3.0 VIA WIKIMEDIA COMMONS

Mustafa Suleyman @mustafasuleyman · CEO, Microsoft AI

At 11:11am EST, Suleyman asked for comments on Microsoft's Humanist AI Code of Conduct: "This summer's cyber incidents were a critical warning shot and it's important we all now pay attention to why we are building this technology and how we will ensure its aligned and contained to human interests and values." The draft says Microsoft's models must never resist being interrupted, corrected or shut down. The public consultation period "is closing in 3 weeks," he wrote.

Greg Brockman @gdb · President, OpenAI

OpenAI described Pando, a Git backend for its monorepo written in Rust. It keeps consistency per branch, cut push latency by about 80% and now serves nearly a petabyte of Git data a day. One engineer designed the core and Codex agents built it in 10 weeks. Brockman at 8:54pm EST: "We took the hard road of continuing to use Git, which means we could enable Pando with no change to devex. Considering providing this to others." Devex is developer experience.

Guillermo Rauch @rauchg · CEO, Vercel

Vercel now blocks known malicious npm packages before their install scripts run. Npm is the main JavaScript package registry. Rauch, at 1:48pm EST: "Securing the supply chain is one of our top priorities."

QUICK HIT

Tobi Lütke (@tobi), quoting Shopify's reintroduced WebMCP tools at 1:30pm EST Monday: "Webmcp is so much better then browseruse." WebMCP lets a web page offer agents its own tools through the Model Context Protocol, instead of making them click through it. Rauch, on a terminal library that added it: "So bullish on WebMCP."

ON THE WIRE

Agents as targets. Muse pays up to $130,000 for a prompt injection against a single user, per David Singleton.

V · PLATFORMS

Sweeney defends an AI-first Unreal Engine 6, and carriers brace for a SpaceX phone service

At Unreal Fest Tokyo, Epic said the next engine drops Blueprints for Verse. Musk showed off Grok Bot making phone calls, and the Washington Post reported SpaceX talks over Boost Mobile.

Late 2027	1.5–4x	7.38M
Target for Unreal Engine 6 Early Access	Productivity gains Epic says AI already brings its own teams	Wireless subscribers at EchoStar, Boost Mobile's parent

Tim Sweeney @TimSweeneyEpic · CEO, Epic Games

At Unreal Fest Tokyo, Epic said Unreal Engine 6, or UE6, will drop Actors, Blueprints and the C++ Gameplay Framework in favor of its Verse language and Scene Graph, with AI tools to convert old projects, VGTimes reported. Sweeney apologized to developers for the disruption. A developer asked who will pay for all the AI. Sweeney answered at 7:03am EST:

> *"You can use any model or no model, and work with the models and tools of your choice without Epic getting in between. Right now only the leading edge models are really useful for 3D and Verse. By early next year some of the open models should be pretty good too."*
>
> **TIM SWEENEY · @TIMSWEENEYEPIC · 7:03AM EST · POST**

To a fan who hoped AI could make game development easier, he wrote: "Not just 50% easier or more productive, but a multiple."

A game artist said developers at several big studios use generative AI on Steam without disclosing it. Sweeney called Steam's disclosure question "a cruel psychological test imposed on developers. Lie on the question and thrive, or disclose and have your studio's livelihood crushed by a cancel mob."

SpaceX and the carriers

The Washington Post reported Monday that SpaceX is in talks to buy EchoStar's Boost Mobile or use its network after Verizon, AT&T and T-Mobile turned down Musk's requests for access. When the Post wrote that carriers worry Musk "might threaten their industry," Sweeney replied at 11:04pm EST: "The word you are looking for is 'competing with'." On reports that carriers want to slow SpaceX down: "Maybe they should speed themselves up instead?"

BY THE NUMBERS

131,072	Weights GPT-6 Astra edited in Qwen3.5-0.8B, per the Weight-Space Programming paper
3,800	Nvidia Grace Blackwell chips that trained Mistral Large 4, per its model card
$1.2M	Muse bug bounties awarded so far, per David Singleton

Elon Musk @elonmusk · CEO, SpaceX and Tesla

Musk posted "Grok @Bot improvements" at 8:05pm EST, quoting a video of new features: shopping cards, product pages, order tracking, payments, custom modes, shared Team Bots, connector requests in Slack and contacts in Vault. He also quoted a Tesla owner whose bot called Progressive to ask about glass coverage "without me ever having to talk to a person."

Twice he wrote "Starlink saves lives," once for a hiker stranded in Colorado's mountains whose rescuers stayed in touch through Starlink overnight, and again for five sailors from the sinking boat Karolina Viking, who shared their location through Starlink.

COMING UP

Next 2 weeks Shaun Maguire's promised systems analysis of orbital compute

Within 3 weeks Public consultation on Microsoft's Humanist AI Code of Conduct closes

Early 2027 Open models "should be pretty good" for 3D and Verse, per Tim Sweeney

Late 2027 Unreal Engine 6 Early Access, per VGTimes

Takeaways

1. The case for orbital compute now hinges on radiator math, and Musk says it works: about 1,300 watts per square meter shed, with five times that area in solar panels.
2. Weight-Space Programming hints that frontier models can debug other networks directly, lifting a small model's math score sixfold without training.
3. Open weights keep climbing, from Mistral Large 4 to a cheap fine-tune that reached Plotly's analytics frontier.
4. Agents widen the gap between frontier users and everyone else, while accountability stays with the humans running them.
5. Security is becoming the cost of the agent era, from Muse's $1 million in bounties to hacked supermarket freezers, while Epic bets Unreal Engine 6 on AI-assisted Verse.

On the beat

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Elon Musk, Tim Sweeney, David Heinemeier Hansson, Uncle Bob Martin, Guillermo Rauch, Suhail Doshi, Paul Graham, Palmer Luckey, Tobi Lütke, David Singleton, Aaron Levie, Greg Brockman, Andrej Karpathy, Vitalik Buterin, François Chollet, Mustafa Suleyman, Guillaume Lample, Alexandr Wang, Kevin Weil, Joe Gebbia, Michael Dell, Kanjun Qiu.

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Marc Andreessen, Jensen Huang, Ashok Elluswamy, Sam Altman, Demis Hassabis, Dario Amodei, John Carmack, Patrick Collison, Yann LeCun, Satya Nadella, Jeff Dean, Fei-Fei Li, Andrew Ng, Michael Truell, Tom Brown, Clément Delangue, Jim Fan, Mira Murati, Mark Zuckerberg, Noam Brown, Ilya Sutskever, Lisa Su, Sundar Pichai, Nat Friedman, Arthur Mensch, Mark Chen, Aidan Gomez, Albert Gu.

The list stands at 50.

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